THE CO-OPERATIVE UNIVERSITY OF KENYA

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LECTURER:	SUSAN JOSIAH
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GROUP MEMBERS:

1. STEPHEN OLOO C024/402205/2023
2. LAMECH MOSEZ C024/401447 /2023
3. BURNINGTONE MWATHA C024//2023
4. SHEM NDARO NGUGI C024/40/2023
5. NELSON LOKOR LOKOR C024/402059/2023

GROUP REPORT

Compiled By: Stephen Oloo

EXECUTIVE SUMMARY

This report explores how IoT is transforming Kenya's agriculture and its role in global value chains. It highlights the dominance of global tech firms in cloud and analytics, creating data dependency risks. A sustainable local model using open-source tools, local partnerships, and smart policies can help Kenyan agritech startups and farmers capture more value and reduce reliance on foreign players.

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Introduction.

Agriculture is a vital part of Kenya's economy, employing over 40% of the population and contributing more than half of the country's GDP directly and indirectly. However, the sector faces major challenges such as climate change, limited irrigation, low access to technology and capital, and global competition. IoT technologies like sensors, drones, and smart irrigation systems present opportunities to boost productivity and efficiency. Kenya is already a leader in digital agriculture, with more than 100 active solutions, yet many IoT components such as devices, platforms, and cloud analytics are produced and managed abroad. This raises concerns about data ownership and value distribution. The report explores these dynamics through four key areas: mapping the global IoT value chain, examining the influence of global tech monopolies, proposing a sustainable local IoT business model, and identifying policy and partnership strategies to strengthen Kenya's position and local innovation capacity.

2.1 IoT in Smart Agriculture

The Internet of Things (IoT) connects devices such as sensors and drones that collect and share data with little human input. In farming, IoT enables precision agriculture through soil and weather monitoring, automated irrigation, and data-driven decisions that improve yields and resource use.

2.2 Smart Agriculture in Kenya

Kenya is well-positioned for smart farming due to strong mobile connectivity and a vibrant agritech scene. It leads Sub-Saharan Africa in digital agriculture solutions, with innovations like SunCulture's solar-powered IoT irrigation and Amini's satellite and sensor data systems. These technologies are improving efficiency, productivity, and sustainability for local farmers.

2.3 Global Value Chain Perspective

IoT in agriculture operates within global value chains, where different parts—such as hardware production, connectivity, cloud hosting, and analytics—are managed across multiple countries. This global setup often places control and profits in the hands of international firms rather than local actors.

2.4 Data, Platform Power and Information Monopolies

The greatest value in digital agriculture lies in data and platforms. Global tech giants like Amazon, Google, Microsoft, and Huawei dominate these layers, creating dependency risks for local players. This concentration of control can lead to data extraction and limit Kenya's ability to benefit fully from its agricultural data.

2.5 Challenges in the Kenyan Context

Key barriers to IoT adoption include high costs, limited technical expertise, poor rural connectivity, and dependence on foreign-made devices and cloud platforms. These factors reduce local ownership and value capture in Kenya's smart agriculture ecosystem.

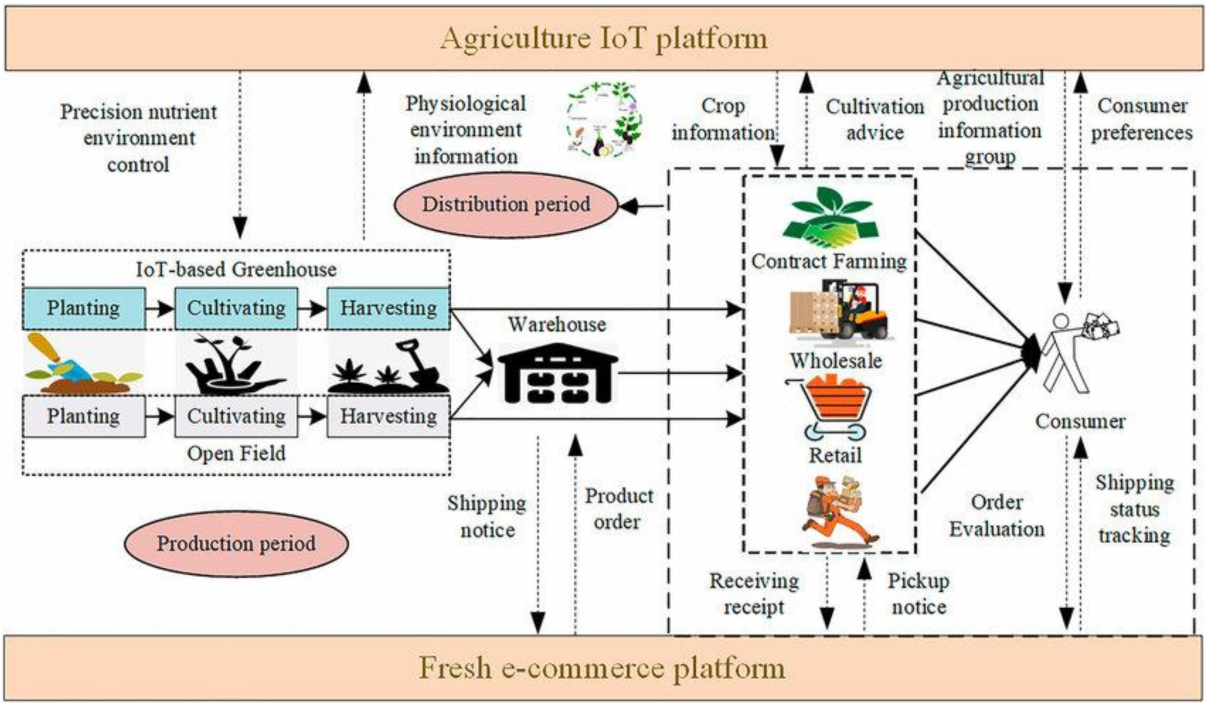
2.6 Opportunities for Kenya

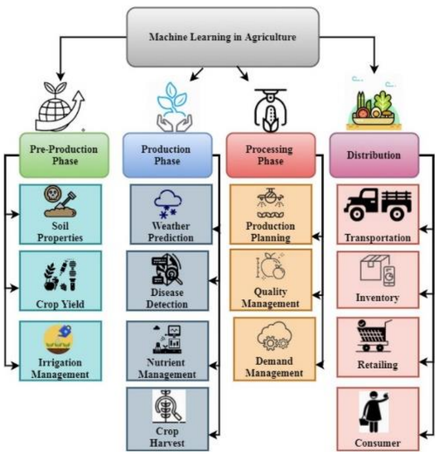
Kenya's growing pool of tech-savvy youth, strong mobile infrastructure, and supportive policy environment create major opportunities for local innovation. By developing context-specific IoT solutions and business models—such as "Irrigation-as-a-Service"—Kenyan firms can build sustainable value and reduce reliance on external technologies.

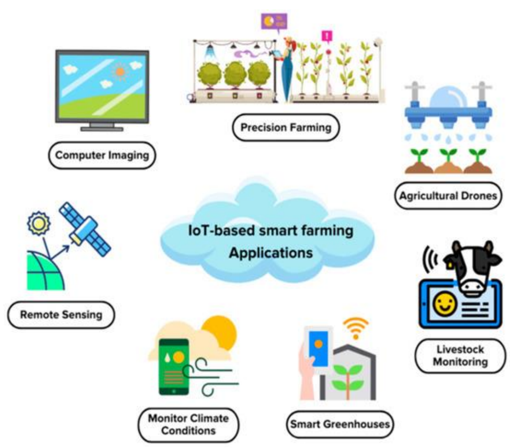
3. Task 1: Mapping the Global IoT Value Chain

3.1 Value Chain Flow

Below is a diagrammatic representation of a typical IoT-enabled smart agriculture solution in Kenya.







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The stages include:

- **Device/hardware manufacturing**: sensors (soil moisture, temperature, humidity), actuators (valves, irrigation controllers), drones. These are often designed and manufactured in China, Europe or the US, with assembly sometimes in low:cost regions.
- **Connectivity/telecommunications**: local connectivity (telco networks in Kenya), LPWAN/5G/LoRaWAN for remote farm areas.
- **Cloud platforms & analytics software**: data collected from devices is sent to the cloud for storage, processing and analytics; this often relies on platforms offered by global tech firms.
- **Applications / services / farm:level interface**: local farmers and agritech service providers access dashboards, mobile apps, alerts and advisory services; they may also receive extension services, data insights, precision:irrigation scheduling.
- **Support & service ecosystem**: local agritech firms, consultants, extension agents, local telcos, equipment installers, financing partners.

3.2 Local vs International Control

A breakdown of who controls which segments in the Kenyan context:

Segment	Typical location/control	Kenyan:local control?
Device design & manufacture	Mostly international (China, US)	Some local assembly/importing only
Connectivity (telco network)	Local telcos in Kenya (e.g., Safaricom, Telkom Kenya)	Strong local presence
Cloud & analytics platform	International providers (Microsoft Azure, AWS, Google Cloud)	Local rollout possible but few local full:stack providers
Applications & services	Local agritech firms & service providers	Strong local relevance
Financing/investment ecosystem	Mix of local and international investors	Growing local startups

For example: An irrigation:as:a:service pilot in rural Kenya used IoT sensors and connectivity but the cloud/analytics infrastructure might still be provided by international platforms.Thus while device/data gathering and farmer interface may be locally managed, value added in manufacturing and analytics often accrues outside Kenya.

3.3 Value Capture Implications

The implication is that although Kenyan farmers benefit from improved yields and services, much of the downstream value (sensor manufacturing profit, platform subscription fees, data monetisation) remains captured by international firms. To shift value capture locally, Kenya must enhance local manufacturing, build local analytics capabilities, and create environments for local data monetisation and service innovation.

4. Task 2: Role of Global Information Monopolies

4.1 Big Tech Involvement

Large technology firms influence the IoT agricultural value chain in several ways:

- They provide cloud infrastructure (e.g., Microsoft Azure, Amazon AWS, Google Cloud) which host IoT platforms and analytics services. Microsoft's "data:driven agriculture" initiative in Africa is an example
- They develop analytics/AI tools that process sensor data and generate insights for farmers.
- They may provide device ecosystems, partner networks, software toolkits and even channel services to agritech companies.
- They shape connectivity infrastructure strategies (edge computing, LPWAN, 5G) and set interoperability or protocol standards. For instance, connectivity models (LPWAN vs 5G) affect cost and data flows.

4.2 Risks for Kenyan Farmers and Businesses

Reliance on global monopolies introduces several risks:

- Data dependency and lock-in
- Data ownership and sovereignty
- Cost and pricing power
- Reduced local innovation
- Vulnerability to external shocks

4.3 Opportunities for Local Players

Despite these risks, there are clear opportunities for local companies such as:

• Local startups can use farmer data to create tailored solutions.
• Local sensor manufacturing can capture more value.
• Partnerships with telcos and cloud firms can localize platforms.
• Locally collected data can drive new analytics services.
• Government policies can support local innovation and growth.

5. Task 3: Proposed Sustainable IoT Business Model for Kenya

5.1 Business Model Description

We propose a Kenyan agritech startup "FamaSense Kenya" that provides a bundled IoT service to smallholder farmers, comprising soil-moisture sensors, smart irrigation control, a mobile app interface, and advisory recommendations.

We will use a hybrid business model that will encompass the following:

- **Subscription fee**: Monthly fee for access to sensors, connectivity, app and analytics.
- **Pay per use component**: Farmers pay per unit of irrigation water or per crop-cycle option.
- **Data monetisation**: Aggregated anonymised data is packaged to agribusinesses or input suppliers (with farmers' consent) to provide insights on soil health, cropping patterns and yield forecasts.

- **Localised manufacturing/assembly**: Basic sensor modules assembled in Kenya (imported components but local assembly) to reduce cost and create local jobs.
- **Open:source analytics layer**: Use an open source analytics stack (or locally hosted cloud like Truehost Kenya) to reduce dependence on global platforms
- **Partnerships**: Partner with local telcos for connectivity, local cooperatives for farmer onboarding, and agribusinesses for input/offtake linkages.

5.2 How it Reduces Dependence on Global Monopolies

- By locally assembling sensors and sourcing where possible regionally, the upstream value stays in Kenya rather than being fully imported.
- By using locally hosted/managed analytics or an open-source software stack, the startup reduces reliance on external cloud giants and limits lockin.
- By aggregating farmer data and providing local insights, the firm retains control of the data value chain rather than ceding it wholly to an external platform.
- By creating a service bundle tailored to Kenyan, the startup competes on local relevance rather than trying to equal large global firms feature by feature

5.3 Revenue & Cost Structure

- **Revenue streams**: Subscription fees; usage fees (irrigation, sensor add:ons); data insight sales; premium services (drone monitoring, crop insurance linkage).
- **Key costs**: Sensor hardware procurement & assembly; connectivity fees; analytics & platform maintenance; farmer onboarding & support; marketing & partnerships.
- **Value proposition**: For farmers – improved yields, reduced input waste, resilience to weather shocks; For agribusinesses – local soil/crop data, supply chain visibility; For government/NGOs – climate:smart agriculture monitoring.
- **Scalability**: Start with clusters of farmers in one region (e.g., around Lake Victoria or Machakos) then expand nationally; sensor modules can be rented or leased to reduce upfront cost.

5.4 Competitive Positioning

- The startup focuses on **affordability**, **local adaptation**, **service bundling** and **data ownership** – differentiating from international firms which may have higher cost, less local adaptation, and data extraction focus.
- Combining hardware, connectivity and advisory as a "one:stop" service lowers complexity for farmers.
- Local partnerships (telco, cooperatives, input suppliers) strengthen network effects and farmer trust.

5.5 Diagram: Business Model Canvas

The diagram below describes the proposed business model canvas

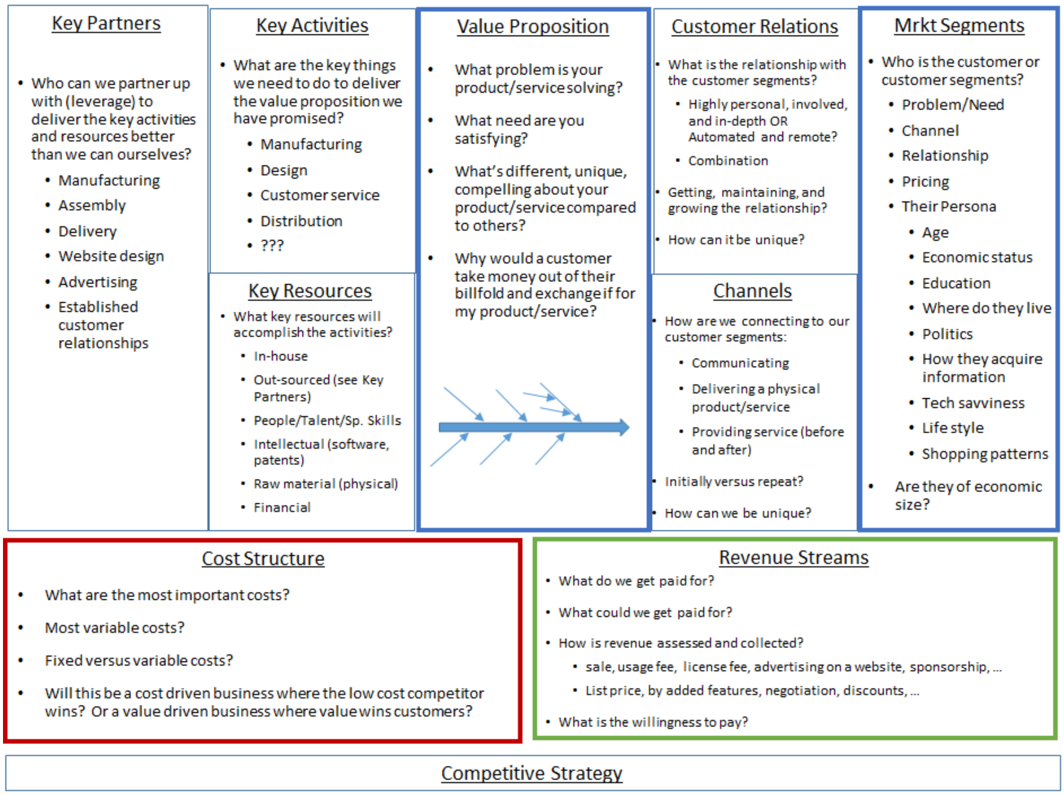

Figure 1: Business Model

6. Task 4: Critical Reflection (Policy, Partnerships, Innovation)

6.1 Policy Enablers

- Kenya should adopt **data sovereignty and data governance** frameworks to ensure that agricultural data generated locally remains accessible and controllable by Kenyan actors. As noted in the context of Kenya's AI sovereignty, infrastructure, governance and human capital are key. (Brookings)
- Incentives for **local manufacturing/assembly** of IoT hardware (sensors, controllers) – e.g., tax breaks, grants, special economic zones (such as Konza Technopolis) (Wikipedia)
- Promote **digital agriculture standards and interoperability** to avoid vendor lock:in and promote local innovation.
- Support **connectivity infrastructure** in rural areas (LPWAN, 5G, affordable internet) so IoT systems can function effectively even in remote farms.
- Create **public:private partnerships** for "Irrigation:as:a:Service" and other shared infrastructure models (e.g., borehole + IoT sensors) as piloted by Mercy Corps in Kenya. (Medium)

6.2 Partnerships & Innovation Ecosystem

- Collaboration between local telcos, sensor providers, agribusinesses and farmer cooperatives to build holistic service models.
- Innovation hubs (such as Konza) and research institutions (universities) can incubate agritech IoT startups, build skills in sensor hardware, data analytics and agronomy.

- Partnerships with NGOs and international donors to subsidise pilots, generate evidence and scale models for smallholders.
- Encouraging open:data platforms (with farmer consent) so aggregated agricultural datasets support local analytics, local service providers and local value capture.

6.3 Shifting Balance of Power

To shift power from global monopolies toward local ecosystems, Kenya must:

- Cultivate **local ownership of data** and platforms rather than just being consumer of foreign:led platforms.
- Enhance **local manufacturing and service capability** so more of the value chain is controlled within Kenya or East Africa.
- Build **skills and human capital** in data science, agronomy, IoT hardware design so startups can innovate rather than rely solely on imports.
- Encourage **regional aggregation and scaling** so Kenyan agritech firms can serve across East Africa and leverage economies of scale.
- Guarantee **transparent, inclusive governance** so smallholder farmers are not sidelined by large firms but can participate in data value share arrangements and benefit equitably.

6.4 Reflection Summary

While Kenya has made significant strides in digital:agriculture and IoT adoption, capturing greater value in the global chain demands deliberate strategy. Without action, Kenyan actors risk being downstream consumers of IoT services, rather than platform owners or value chain leaders. The recommended combination of policy, partnerships, innovation ecosystems and business model design offers a pathway for Kenya to move from adoption to ownership and value capture.

7. Conclusion & Recommendations

This report has shown that IoT technologies hold significant promise for transforming Kenyan agriculture through yield improvement, resource efficiency and integration into global value chains. However, the value chain of IoT is internationally distributed—device manufacturing, cloud analytics and platform control often reside outside Kenya—raising risks of dependency and reduced local value capture.

Key recommendations:

1. **Promote local manufacturing/assembly of IoT hardware** to capture upstream value.
2. **Adopt data governance and sovereignty policies** to ensure Kenyan actors retain access to and control of agricultural data.
3. **Support agritech business models that bundle hardware + connectivity + advisory services**, targeted at smallholder farmers, with revenue models such as subscription/pay:per:use/data monetisation.
4. **Strengthen connectivity infrastructure in rural areas**, including LPWAN, 5G and hybrid models to enable reliable IoT services.
5. **Encourage partnerships** between telcos, agritech startups, cooperatives, extension services and research institutions to build local ecosystems.

6. **Invest in human capital and innovation hubs** so Kenya develops capability in sensor design, data analytics and agronomy for IoT farming.
7. **Facilitate regional scaling** so Kenyan solutions can reach East African markets, increasing economies of scale and bargaining power.

By implementing these actions, Kenya can shift from being a consumer of IoT:enabled agriculture services to a producer of value, capturing more of the benefits of the global IoT value chain and supporting inclusive, climate:smart agriculture for its farmers.

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Appendices

Appendix A: IoT Value Chain Diagram

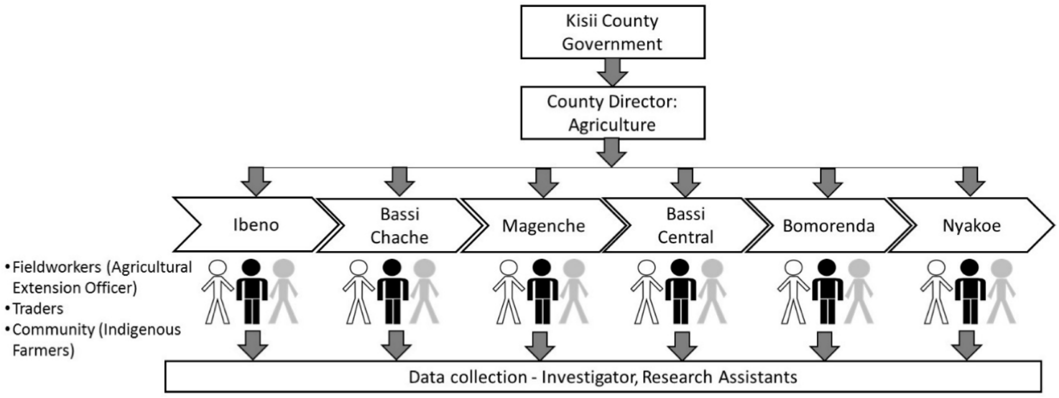

Figure: IoT value chain

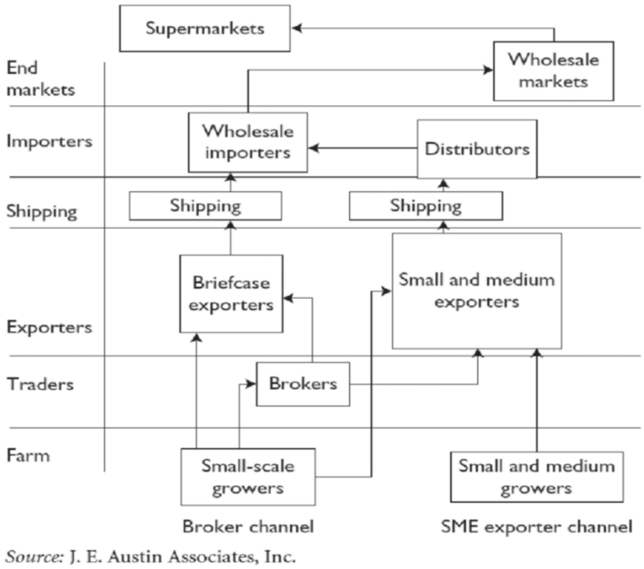

Figure: IoT Value Chain 2